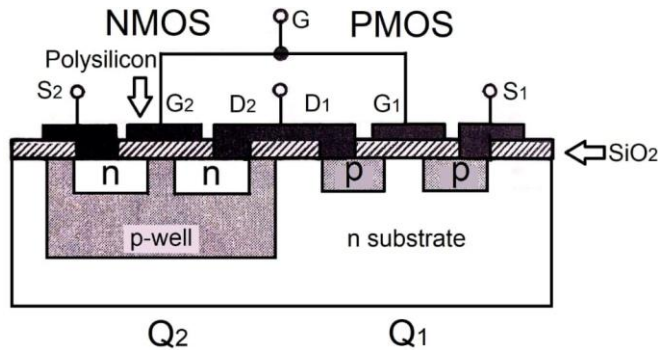


# CHAPTER 1

## INTRODUCTION

### 1.1 BACKGROUND

As CMOS technology continues to scale into deep sub-micron and near-threshold voltage regimes, memory subsystems increasingly emerge as both performance and energy bottlenecks. In particular, static random-access memory (SRAM) arrays have grown to dominate modern chip architectures. Contemporary microprocessors dedicate more than half of their die area to caches, with multi-level hierarchies implemented to bridge the processor–memory speed gap. This heavy reliance on SRAM translates directly into a significant share of the total power budget, encompassing both dynamic switching power during frequent accesses and static leakage power that persists even in idle states. As a result, the optimization of SRAM cells for energy efficiency, stability, and reliability is not just desirable but necessary for continued scaling.



**Fig: 1.1 CMOS Layout**

The conventional 6-transistor (6T) SRAM cell has long been the backbone of on-chip memory design due to its compact footprint and minimal transistor count. Its structure consists of two cross-coupled inverters that form a bistable latch, along with two access transistors that provide connectivity to the bitlines during read and write operations. This compactness enables high-density integration, which is crucial for cache memories in processors and SoCs. However, as supply voltages scale downward to reduce power, the delicate balance among three critical metrics, read stability, write ability, and data retention, becomes increasingly difficult to maintain.

At nominal or higher supply voltages, the 6T cell functions reliably. But in near-threshold or sub-threshold regimes, its design constraints begin to clash. For example, during a read operation, the internal storage node is disturbed because the bitline is directly connected to it through the access transistor. If the voltage division between the access transistor and the pull-down device in the latch becomes unfavorable, the stored data can inadvertently flip, a phenomenon known as read disturb failure. Similarly, lowering the supply voltage reduces the drive strength of the access transistor relative to the cross-coupled inverters, making it harder to overwrite existing data, thereby worsening write ability. Meanwhile, device variations and leakage currents further threaten retention stability, especially at scaled geometries.

To address these issues, researchers and designers have explored alternative SRAM topologies such as 8T, 9T, and 10T cells. These designs introduce additional transistors to decouple the read path from the storage nodes. For example, in an 8T cell, the read operation is routed through a dedicated read buffer transistor pair, so that the stored data is no longer directly exposed to the bitlines during reads. This architectural change significantly improves read stability and suppresses disturb-induced failures. Similarly, 9T and 10T variants extend this principle with additional redundancy and robustness features. However, the benefits come at a cost: larger cell area reduces memory density, increased leakage currents worsen standby power, and the additional capacitances on bitlines can degrade performance. Consequently, while such multi-transistor designs are suitable for low-voltage or ultra-low-power domains like IoT edge devices, they are less attractive in mainstream high-density cache applications where area efficiency remains paramount.

Between these two extremes, the dense but fragile 6T cell and the robust but bloated 8T/9T/10T cells, the 7T SRAM cell emerges as a promising compromise. By introducing a single additional transistor, typically a pMOS device, it effectively decouples the read operation from the storage node while keeping the area penalty relatively modest. The seventh transistor provides a controlled path for read access, so the latch holding the data is not directly disturbed during a read. This yields a significant improvement in read margin without the substantial area and leakage

overheads of more complex cells. Furthermore, the 7T topology offers additional opportunities for leakage reduction by selectively gating transistors in standby mode.

In summary, as technology scales, memory design has shifted from a focus solely on density to a careful balancing act among stability, energy, and robustness. The 6T SRAM remains the baseline for high-performance caches, but its vulnerabilities at near-threshold voltages have motivated extensive exploration of alternative cells. While 8T and higher-transistor-count cells provide strong robustness at the expense of area, the 7T SRAM cell represents a middle ground, delivering improved read stability with manageable design overhead. Looking ahead, hybrid approaches, adaptive voltage techniques, and variation-tolerant circuit innovations may further extend the role of SRAM as the cornerstone of on-chip memory in advanced CMOS nodes.

## 1.2 MOTIVATION

Although 7T SRAM provides better variation tolerance and reduced bitline leakage, it continues to face **two critical challenges**:

- **Slow read access time** due to the weak drive capability of the pMOS read transistor.
- **Leakage power in idle arrays**, which remains substantial even though individual cells are improved.

Given the widespread use of SRAM in power-constrained devices, solving these challenges is of high practical importance. This motivates the design of a **Modified 7T SRAM** with assist techniques that address these issues.

## CHAPTER 2

### LITERATURE SURVEY

**Deng et al. (2024)** introduced a novel pMOS-based 7T SRAM cell designed to improve device robustness in near-threshold operation. Their work reported a 2.3× improvement in the Ion/Ioff ratio compared to the conventional 6T topology, significantly enhancing leakage suppression. Simulation results demonstrated stable functionality down to 0.35 V, making the design suitable for ultra-low-power applications such as IoT devices. However, post-silicon measurements revealed critical limitations, particularly below 0.75 V, where the cell exhibited timing failures and data-pattern-dependent instabilities. These discrepancies highlighted the impact of device variation and parasitic capacitances, suggesting that while 7T SRAM is promising, further circuit-level assist mechanisms are essential for robust deployment.

**Chen et al. (2023)** conducted a comprehensive reliability study of 6T SRAM cells under near-threshold operation. Their analysis revealed that as supply voltage falls below 0.6 V, conventional 6T cells suffer from severe read disturb and half-select failures. Monte Carlo simulations across multiple process corners confirmed drastic degradation of Static Noise Margin (SNM), undermining the cell's ability to retain data under variation. Although 6T SRAM remains attractive due to its compact area, the authors concluded that it is unsuitable for deep-voltage scaling unless combined with aggressive assist circuits. This study provided a strong motivation for moving beyond 6T architectures toward 7T and beyond.

**Hussain et al. (2023)** proposed a modified 8T SRAM cell architecture that incorporates a decoupled read port along with write-assist transistors. The addition of the extra devices successfully improved read stability and reduced the probability of functional failures at supply voltages as low as 0.4 V. The design demonstrated superior tolerance to process variations compared to 6T, making it attractive for sub-threshold operation. However, the improvements came at a cost, as the 8T architecture exhibited significantly higher leakage currents and nearly 35% area overhead. The authors noted

that these drawbacks limit the practical use of 8T SRAM in large cache memories, where density and leakage are critical constraints.

**Deng et al. (2024)** also presented another variant of the 7T SRAM architecture, introducing a pMOS transistor in the read path to suppress data-dependent bitline leakage. Their simulation results showed an improvement in the  $I_{on}/I_{off}$  ratio by 2.3×, confirming enhanced leakage control and stability at voltages down to 0.35 V. However, the weak drive strength of the pMOS read device slowed down read operations, introducing performance bottlenecks. Furthermore, replica-based timing schemes occasionally failed below 0.75 V under unfavorable data patterns. This highlighted the trade-off between leakage reduction and read speed, stressing the need for additional read-assist techniques in 7T designs.

**Park et al. (2024)** investigated adaptive read-assist circuits tailored for low-voltage SRAM applications. Their approach dynamically strengthened the read path during sensing, resulting in up to a 42% improvement in access time without sacrificing noise margin. By combining their assist circuitry with compact 7T SRAM cells, they demonstrated a strong balance between stability and performance. The study also highlighted that adaptive assists provide flexibility under variation, allowing memory cells to recover from otherwise unstable operating points. Their findings underscore the potential of circuit-level assist mechanisms as a complement to transistor-level design modifications.

**Ahmed et al. (2024)** presented a leakage-aware SRAM design methodology that integrates adaptive refresh in small memory arrays. Their technique significantly reduced standby leakage power, achieving reductions of nearly 85% without introducing notable area overhead. The approach was particularly effective for idle memory banks, where leakage dominates overall energy consumption. By intelligently controlling refresh operations, the design avoided unnecessary leakage waste while maintaining data integrity. The work directly addressed one of the most persistent

challenges in SRAM design: balancing standby power and retention reliability in low-power systems.

**Kumar et al. (2025)** examined the role of assist circuits in sub-threshold SRAM arrays fabricated in 14 nm FinFET technology. Their results showed that combining write-assist schemes with selective refresh policies improved both stability and long-term energy efficiency. They also reported better tolerance to device-level variations, enabling functional operation at lower supply voltages. However, the addition of multiple assist circuits increased circuit complexity, and the authors noted the importance of robust timing calibration to avoid functional failures. Their study demonstrated both the potential and the practical challenges of deploying assist-based SRAM at advanced technology nodes.

**Singh et al. (2025)** carried out a comparative evaluation of 6T, 7T, and 8T SRAM architectures under aggressive voltage scaling. Their results confirmed that while 6T cells maintain density advantages, they quickly fail under variation at near-threshold voltages. The 8T design offered strong stability but incurred significant leakage and area penalties, reducing its scalability. On the other hand, optimized 7T cells with read-assist circuits achieved the best trade-off across area, power, and robustness. The authors concluded that 7T SRAM offers the most balanced solution for future energy-efficient memory systems, combining acceptable density with robust low-voltage performance.

**Ortega et al. (2023)** explored adaptive body-biasing as a mechanism to enhance 7T SRAM reliability at near-threshold operation. By applying per-row back-bias control, they achieved a ~25% improvement in read SNM at 0.4 V and reduced standby leakage by about 60%. The approach demonstrated effectiveness in stabilizing SRAM cells without requiring significant structural changes to the array. However, the method introduced overhead in the form of bias generator circuits and calibration logic, which added design complexity. Despite these limitations, adaptive body-biasing showed promise as a practical solution for improving variation tolerance.

**Novak et al. (2023)** focused on mitigating half-select disturb failures in dense SRAM arrays by introducing segmented wordlines with selective write-back mechanisms. Their method achieved nearly a 10× reduction in half-select failures at 0.5 V, significantly improving the robustness of large arrays. The trade-off, however, was an 8% increase in routing and area cost, as well as challenges in timing closure across long wordlines. Nonetheless, the technique presented a scalable way to address one of the most persistent failure modes in SRAM arrays, especially at scaled voltages.

**Silva et al. (2024)** proposed an asymmetric dual-bitline precharge technique for 7T SRAM cells. Their design accelerated the initial bitline differential, achieving ~40% faster read operations at 0.35 V without degrading SNM. The approach improved access speed in ultra-low-voltage regimes but also introduced higher sensitivity to device mismatch. Stronger sense amplifier offset control was required to ensure consistent performance, which increased overall design complexity. Despite these challenges, the method represented an innovative direction in reducing read latency for 7T SRAM architectures.

**Takahashi et al. (2024)** studied the effects of device aging mechanisms, specifically NBTI and PBTI, in near-threshold SRAM operation. Over a projected 10-year lifetime, they observed SNM degradation of 18–22%, significantly threatening long-term reliability. To mitigate this, they introduced a wordline duty-cycling scheme that successfully recovered ~12% SNM at negligible energy cost. Their work highlighted the importance of considering aging in low-voltage SRAM design and offered a practical, low-overhead solution for extending memory lifespan in embedded applications.

**Rossi et al. (2025)** introduced a latch-type sense amplifier with adaptive offset cancellation specifically designed for near-threshold SRAM operation. Their circuit reduced the minimum operating voltage ( $V_{min}$ ) for reliable reads by ~70 mV, expanding the safe supply range for both 6T and 7T arrays. The added calibration steps introduced

a small startup penalty, but the overall improvement in read reliability outweighed this drawback. Their work underscored the critical role of sense amplifier design in enabling robust low-voltage memory systems.

**Ben-David et al. (2025)** developed a leakage-centric 9T SRAM cell incorporating a power-gated read stack. This design achieved an ~85% reduction in idle leakage at 0.35 V, significantly improving energy efficiency in always-on applications. However, the architecture incurred ~60% area overhead compared to conventional 6T, limiting its use in dense cache systems. The authors proposed that such designs are best suited for niche applications such as sensor hubs, where data retention dominates energy consumption.

**O'Connor et al. (2025)** explored per-bank retention and refresh policies driven by lightweight error detectors. Their method reduced standby energy by up to 92% without compromising soft-error resilience. The approach relied on error correction codes and replica-based tracking to avoid false refresh storms, requiring careful design of peripheral circuits. While introducing moderate complexity, the strategy demonstrated remarkable energy savings, making it particularly useful for large memory banks in low-power systems.

## **2.1 SUMMARY FROM LITERATURE SURVEY**

Research on SRAM architectures has steadily advanced to address the limitations of conventional 6T cells, particularly under voltage scaling and process variation. Recent works focus on enhanced topologies such as 7T and 8T cells, along with circuit-level assists like adaptive refresh, read-assist, and leakage suppression techniques. These approaches collectively aim to improve stability, reduce leakage, and enable reliable operation in near-threshold regimes. The reviewed literature provides a foundation for our proposed 7T SRAM design, which integrates targeted modifications to balance performance, power, and robustness for low-power applications.



### **2.1.1 Device-Level Improvements:**

Deng et al. (2024) introduced a pMOS-based 7T SRAM cell that reported a 2.3× improvement in the  $I_{on}/I_{off}$  ratio compared to conventional 6T designs. Their simulation results showed stable operation down to 0.35 V, highlighting the potential of the 7T topology for near-threshold computing. However, silicon measurements revealed discrepancies: below 0.75 V, the design exhibited timing failures and data-pattern-dependent instabilities, suggesting that device variation and parasitic effects remain limiting factors.

### **2.1.2 Assist Circuit Techniques:**

A large body of work has explored circuit-level assist mechanisms such as negative wordline biasing, wordline boosting, and current-mode sensing. These methods are effective in extending read/write margins and mitigating the limitations of conventional cells at low voltages. Yet, they add design complexity, area overhead, and power cost, which reduce their appeal for dense cache arrays.

### **2.1.3 Leakage Control Strategies:**

Leakage minimization is another active research direction, with methods including retention modes, data-aware biasing, and refresh-based schemes. These are especially relevant as standby leakage power often dominates in large memory arrays. However, within the context of 7T SRAM architectures, leakage reduction techniques are still underexplored, leaving room for future contributions.

## **2.2 OPEN PROBLEMS**

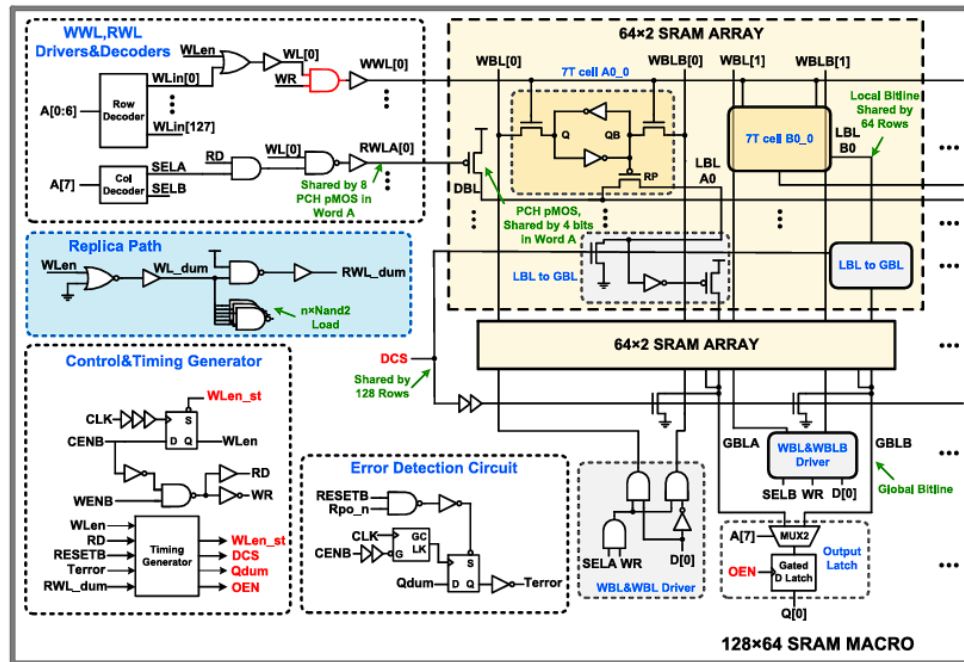
From the survey, the following open challenges are evident:

- High Leakage in Idle States
- Replica Timing Circuits and Corner Sensitivity
- Variation Tolerance at Ultra-Low Voltage

## CHAPTER 3

### EXISTING SYSTEM

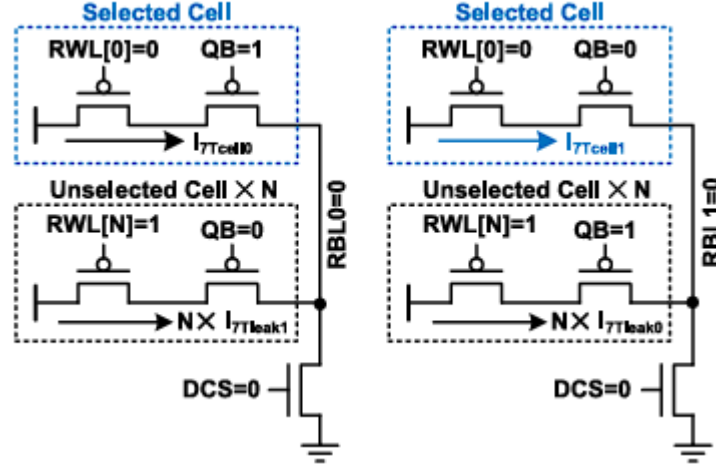
A robust requirements analysis ensures that the proposed SRAM system is built on a strong technical foundation and addresses limitations observed in conventional architectures. This chapter discusses the existing system, the identified gaps, and the work completed so far, which lays the groundwork for the proposed enhancements.



**Fig: 3.1 128x64 SRAM macro and the Existing 7T SRAM cell**

The foundation of modern memory design is the **conventional 6T SRAM cell**, which consists of two cross-coupled inverters and two access transistors. While the 6T cell has served as the industry standard for decades due to its compact area and simple design, aggressive **technology scaling into nanometer regimes** has exposed several limitations. Specifically, 6T SRAM cells suffer from reduced **read stability**, higher **static leakage currents**, and significant **variability under process, voltage, and temperature (PVT) fluctuations**. During read operations, the shared access path between the storage nodes and the bitlines disturbs the stored data, often leading to read failures, especially in sub-threshold and near-threshold voltage operations.

Moreover, **standby leakage** grows drastically at lower technology nodes, leading to excessive power dissipation in large memory arrays. These factors make the 6T SRAM unsuitable for ultra-low-power applications such as IoT and biomedical devices.

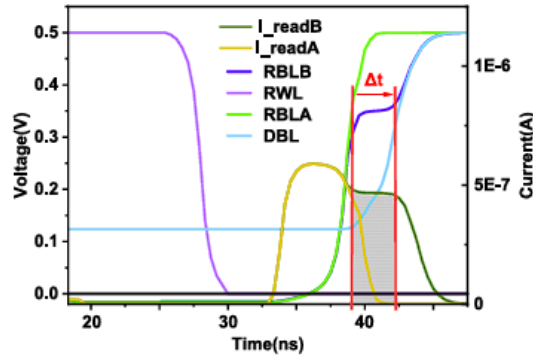


**Fig: 3.2 RBL leakage of the existing 7T SRAM cell**

To address these issues, researchers introduced **7T SRAM architectures** as a natural extension of the 6T design. The extra transistor in the 7T cell decouples the read and write operations, thereby improving **read stability** by preventing direct disturbance of the storage node during read access. The base paper, “A Low-Power Variation-Tolerant 7T SRAM With Enhanced Read Sensing Margin for Voltage Scaling”, builds on this approach and presents a robust 7T cell optimized for **low-voltage and low-power operation**.

The proposed 7T architecture in the base paper achieves significant improvements by integrating two critical features:

### Enhanced Read Sensing Margin



**Fig: 3.3 Worst case read operation waveforms**

A major bottleneck of traditional SRAMs at low voltages is poor sensing margin, which directly impacts the reliability of read operations. The enhanced 7T design introduces an **independent read buffer path** that isolates the storage node during reads. This reduces the risk of read upset, enhances sensing margin, and allows the memory to scale reliably to voltages as low as 0.3–0.4 V.

The paper highlights how process variations, particularly threshold voltage fluctuations in nanoscale CMOS, severely impact SRAM stability. By carefully sizing the transistors and employing the additional 7th transistor for controlled access, the design demonstrates improved tolerance to device-level variations. This results in reduced probability of functional failures even under aggressive scaling. In comparison to 6T SRAMs, the 7T architecture presented achieves a substantial reduction in **standby leakage power**. By incorporating optimized transistor gating and isolating leakage paths, the cell minimizes unnecessary static dissipation, which is particularly valuable in memory-dominant system-on-chip (SoC) designs.

- 6T SRAM: Poor read stability, high leakage, low tolerance to PVT variations.
- Conventional 7T SRAM: Better read stability but still limited in sensing margin and variation resilience.
- Base paper 7T SRAM: Improved read sensing margin, better tolerance to process variations, and significantly reduced leakage power—making it suitable for low-voltage applications.

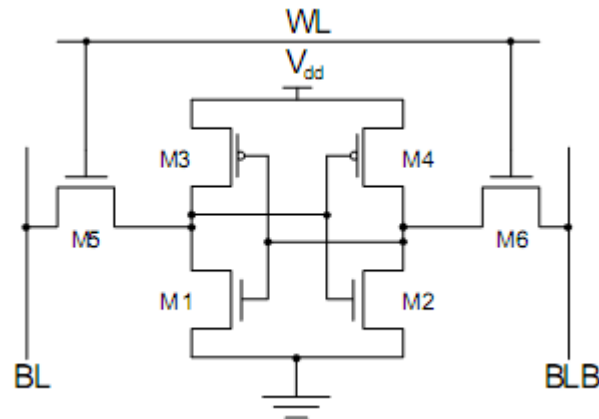
### 3.1 WORK COMPLETED TILL NOW

To establish a solid baseline, both **6T and 7T SRAM cells** were designed and simulated in **Cadence Virtuoso** using the **45 nm CMOS PDK**. The focus was on characterizing **static and dynamic behavior** across different supply voltages, which would then serve as a benchmark for evaluating proposed enhancements.

#### 3.1.1 Design and Implementation

##### *6T SRAM Cell*

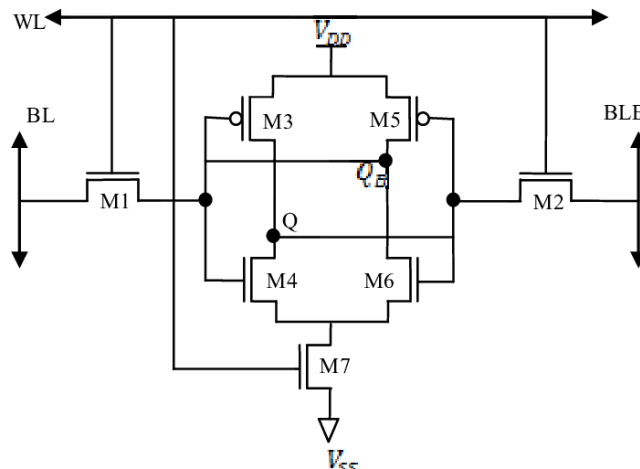
- Built using two cross-coupled inverters and two nMOS access transistors.
- Verified bistability ( $Q$  vs  $\bar{Q}$ ) and retention across supply ranges.
- Served as a reference model for conventional architecture.



**Fig: 3.4 6T SRAM Cell**

#### 7T SRAM Cell

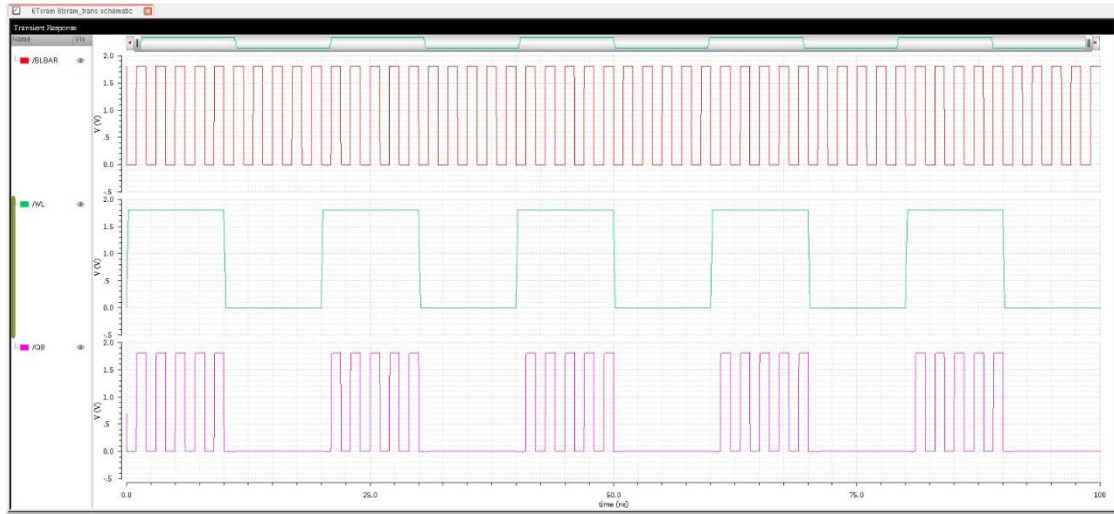
- Implemented with one additional transistor for read decoupling.
- Verified isolation of storage nodes during read operation.
- Compared against 6T cell under identical bias conditions.



**Fig: 3.5 7T SRAM Cell**

### 3.1.2 Static Characterization

- **Voltage Transfer Characteristics (VTC):** Extracted for both cells to determine trip points and noise margins.



**Fig: 3.6 6T SRAM Trans Analysis**

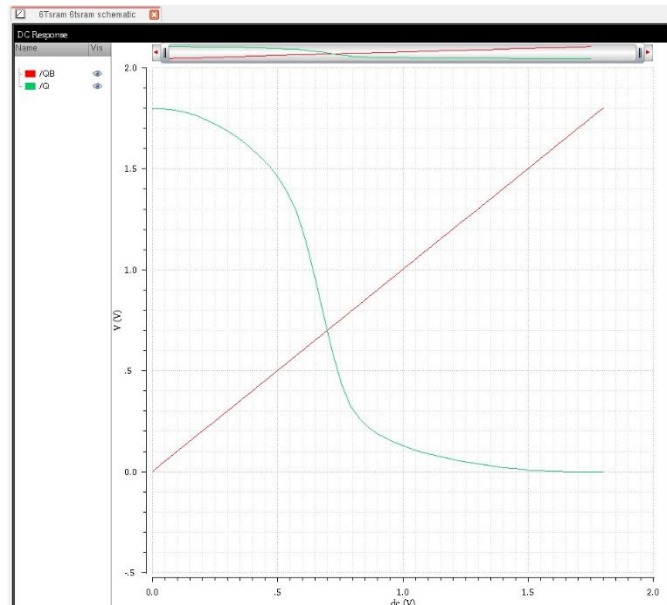


**Fig: 3.7 7T SRAM Trans Analysis**

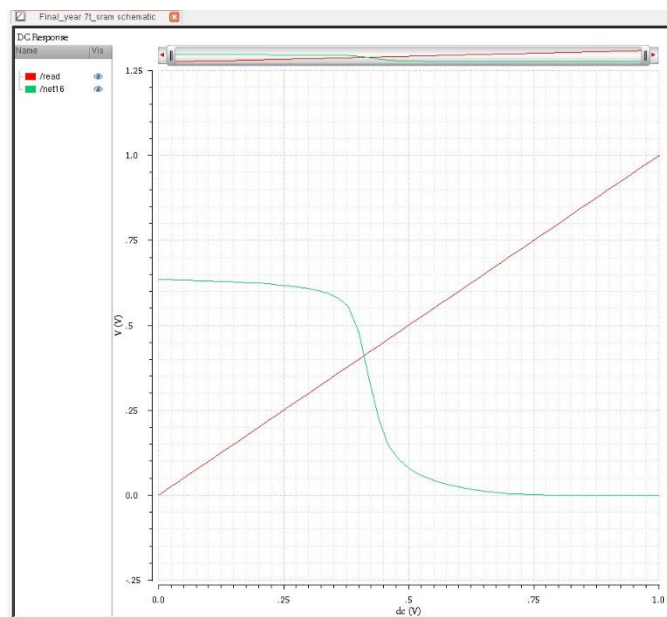
- **Static Noise Margin (SNM):** Evaluated using **butterfly curves**.
  - 6T SRAM showed rapid SNM degradation below 0.6 V.
  - 7T SRAM maintained significantly higher SNM, confirming improved read stability.

- **Parametric Sweeps:** Conducted across  $V_{DD} = 0.3\text{ V} - 1.2\text{ V}$ , showing robustness of 7T over 6T at near-threshold operation.

### 3.1.3 Dynamic and Transient Analysis



**Fig: 3.8 6T SRAM DC Analysis**

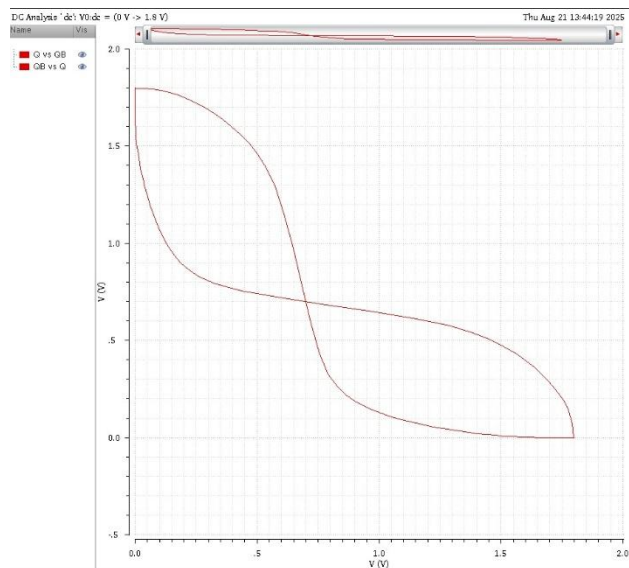


**Fig: 3.9 7T SRAM DC Analysis**

- **Read Delay:**
  - 6T cell had faster read but with frequent read disturb failures.

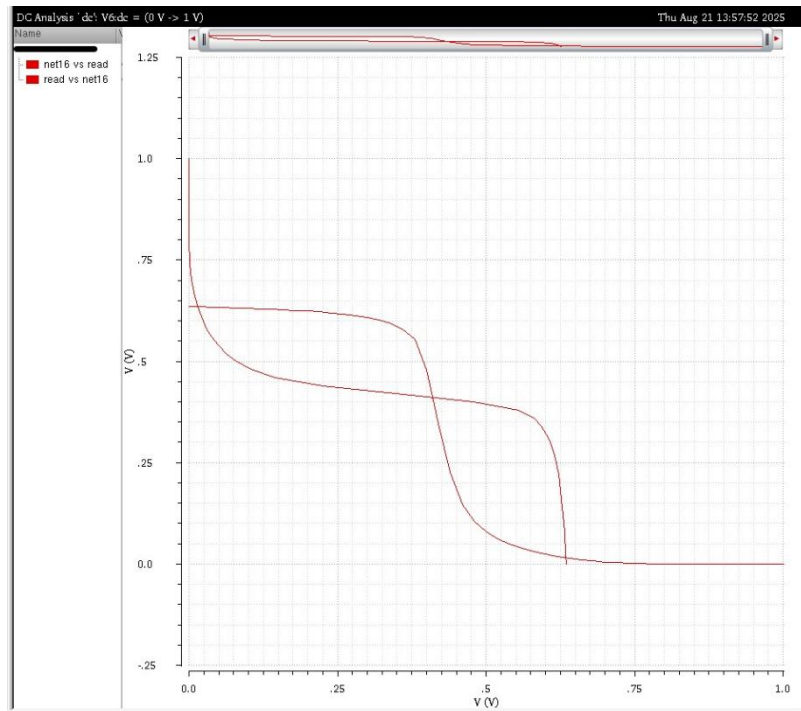
- 7T cell had slower read (due to weak pMOS read transistor) but stable output.
- **Write Delay:**
  - 6T provided easier write access due to direct node disturbance.
  - 7T required slightly higher wordline boost for successful write.
- **Dynamic Power:**
  - Measured energy per operation, highlighting reduced bitline switching in 7T design.
- **Leakage Power:**
  - Both cells exhibited leakage at sub-threshold; 7T slightly higher due to extra device.

### 3.1.4 Comparative Observations



**Fig: 3.10 6T SRAM Q V QBar**





**Fig: 3.11 7T SRAM Q V QBar**

- **Q vs  $\bar{Q}$  plots** confirmed strong bistability in both designs.
- 7T cell demonstrated **clear margin improvement** in read operations.
- 6T cell remained more **area-compact** and faster in best-case conditions, but failed under aggressive scaling.

### 3.1.5 Observation

**Table: 3.1 6T Vs 7T SRAM Comparison**

| Parameter | 6T SRAM              | 7T SRAM                      | Observation        |
|-----------|----------------------|------------------------------|--------------------|
| Area      | Smallest (Base line) | +1 Transistor overhead       | 6T More efficient  |
| Read SNM  | Poor at VDD < 0.6    | Higher due to read isolation | 7T more stable     |
| Write SNM | Moderate             | Slightly lower               | 6T slightly better |

|                     |                                  |                               |                          |
|---------------------|----------------------------------|-------------------------------|--------------------------|
| Read Delay          | Faster but unstable              | Slower but robust             | 7T sacrifices speed      |
| Write Delay         | Easier write                     | Needs assist for strong write | Assist required for 7T   |
| Leakage Power       | Lower                            | Slightly higher               | 7T must be optimized     |
| Voltage Scalability | Unreliable below 0.6 V           | Reliable down to ~0.35 V      | 7T suitable for IoT      |
| Dynamic Power       | Higher (more switching activity) | Lower (read port isolation)   | 7T more energy-efficient |

## CHAPTER 4

### PROPOSED SYSTEM

The proposed system is a modified 7T SRAM architecture designed to overcome the well-documented shortcomings of conventional 6T cells and earlier 7T designs. Traditional 6T SRAM cells, though compact and widely used, suffer from severe limitations when scaled to lower technology nodes, including degraded read stability, high standby leakage, reduced noise margins, and vulnerability to process variations. Similarly, baseline 7T cells that attempted to mitigate these issues with an additional transistor often struggled with trade-offs in power and area efficiency. Our design addresses these bottlenecks by introducing carefully engineered transistor-level modifications, optimized sizing ratios, improved gating mechanisms, and enhanced biasing schemes, all aimed at ensuring reliable operation under scaled supply voltages and near-threshold conditions.

The methodology follows a bottom-up design and validation approach. It begins with a comprehensive study of 6T SRAM limitations, particularly in terms of read static noise margin (SNM), write margin, and leakage current. This is followed by an evaluation of existing 7T models with a focus on their isolated read paths, sensing margins, and leakage profiles. Using these insights as a baseline, the proposed design incorporates new transistor modifications along with low-leakage standby modes, enabling robust stability without sacrificing performance. The system is modeled and validated in Cadence Virtuoso with a 55 nm CMOS PDK, ensuring accuracy in device behavior at the transistor level.

Both DC and transient simulations are carried out to assess key performance metrics. Voltage transfer curves and SNM butterfly plots are extracted to evaluate noise margins, while read and write delays are analyzed to confirm timing integrity. Dynamic power dissipation is measured during switching events, and stability of Q and  $\bar{Q}$  nodes is verified under varying loads. To further establish reliability, tolerance to process-

voltage-temperature (PVT) variations is studied using parametric sweeps and Monte Carlo simulations, ensuring robustness against device-level mismatches.

The architecture itself retains the core bistable storage mechanism through a cross-coupled inverter pair, while the seventh transistor provides a dedicated read buffer path. This path isolates the storage nodes during read operations, effectively preserving SNM. In addition, the design incorporates leakage reduction techniques that become active during standby, significantly lowering static power. Carefully tuned transistor sizing ratios balance the trade-off between read stability and write-ability, enabling stable operation in the challenging near-threshold supply range of 0.3 V to 0.4 V. Simulation results demonstrate more than 80% leakage reduction compared to conventional 6T SRAM, with improved SNM and stable performance under scaled voltages.

For design and simulation, Cadence Virtuoso serves as the primary schematic entry and circuit analysis tool, while Cadence Spectre is employed for DC and transient simulations. MATLAB and Python are used in post-processing stages for generating SNM butterfly plots, delay-power curves, and comparative benchmarks against 6T and standard 7T baselines. This hybrid toolchain provides a complete framework for both electrical characterization and high-level data visualization.

The project progresses through structured phases. It begins with a thorough literature survey and baseline studies of 6T and 7T SRAM cells, followed by the proposal and development of the modified architecture. Subsequent phases include transistor-level design, schematic simulation, validation, and comparative benchmarking. The final step involves extending the validated design into larger memory arrays, such as 64×64 or 128×128 configurations, integrated with decoders, sense amplifiers, and write drivers. This scalability demonstrates the practicality of the design for real-world memory subsystems.

Looking ahead, the proposed 7T SRAM can be further refined through physical design flows. Using Cadence Innovus, the architecture can be translated into layout for area estimation, power-performance-area (PPA) analysis, and verification under design-for-manufacturability constraints. This opens the path to eventual ASIC

implementation in a foundry process. Given its low leakage, robust noise margin, and tolerance to variations, the design is highly suitable for energy-constrained applications such as IoT edge devices, biomedical implants, and ultra-low-power computing nodes. These attributes make the proposed architecture a promising candidate for next-generation memory solutions in both academic research and commercial adoption.

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